Radiology Order-to-Report Workflow: Case Study

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This case study maps how a radiology order travels from the EMR to a signed report, defines the requirements and database behind it, and measures where time is lost using synthetic data.

Problem and goal

Radiology reports reach the ordering doctor later than they should, and nobody can see which step caused the delay. An order passes through several systems (EMR/HIS, RIS, the imaging device, PACS) and back to the EMR, and a delay or a data error at any handoff is easy to miss.

The goal is to make that path visible: map it, define what each system must send, store every status change in a database, and measure the time spent in each stage.

The project delivers a workflow and data-flow map, a requirements table, a PostgreSQL schema, a synthetic-data generator, KPI queries, and a dashboard.

Scope and stakeholders

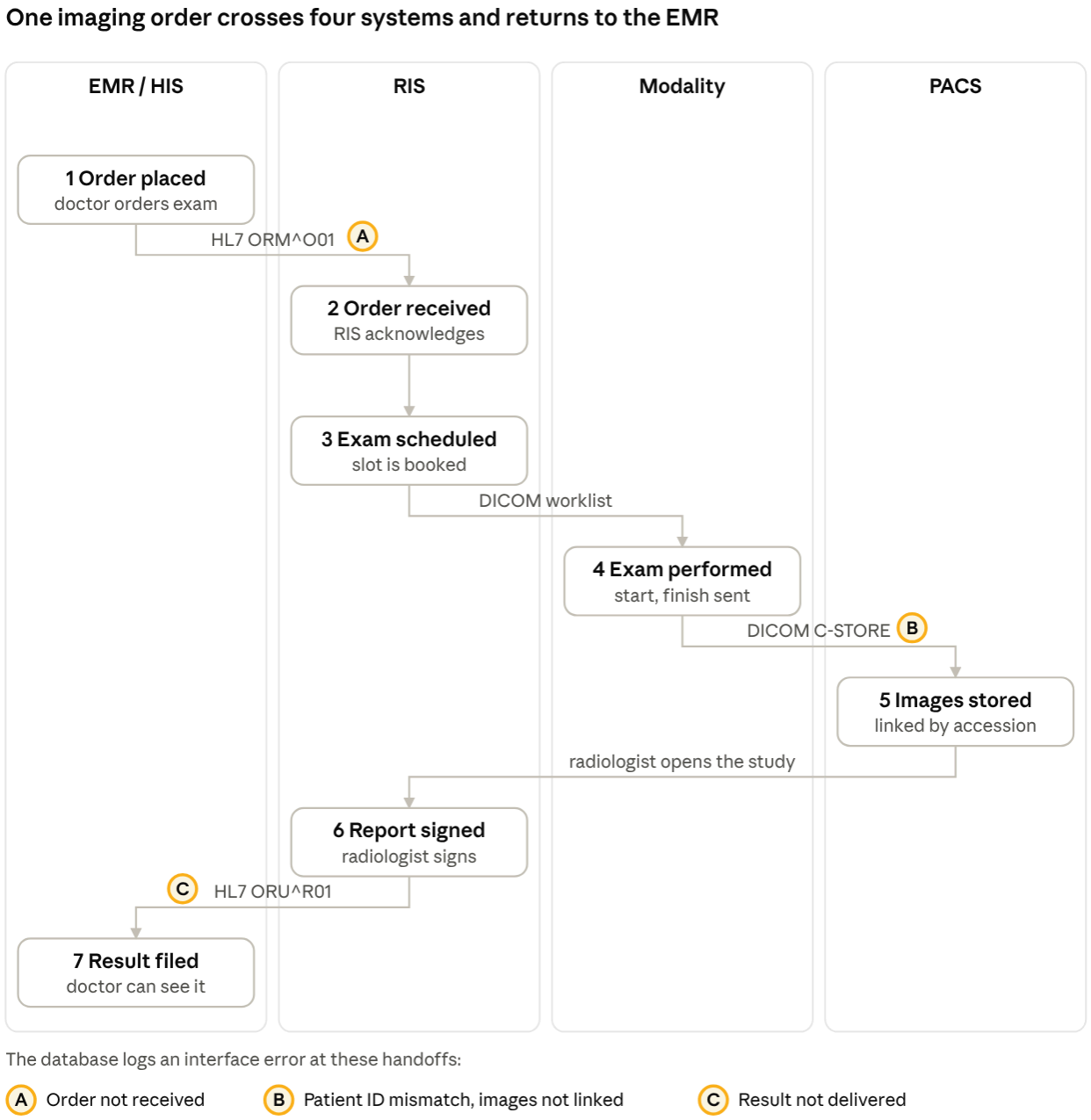
The scope is imaging orders (X-ray, CT, MRI) from the moment a doctor places the order to the moment the signed report is visible in the EMR. Billing, image storage internals and AI reading are out of scope.

Role	What they need from the workflow
Ordering doctor	An order that reaches radiology complete the first time, and the signed report back in the EMR quickly
Scheduler	The right exam code, priority and patient identity, so a slot can be booked without phoning the ward
Radiographer	The exam on the modality worklist with correct patient data, and a way to mark it started and completed
Radiologist	Images in PACS linked to the right order, the clinical history, and a queue sorted by priority

Role	What they need from the workflow
Hospital IT	HL7 and DICOM interfaces that log failures and can be traced message by message
Radiology manager	Turnaround by stage, priority and shift, to decide staffing and fix bottlenecks

These needs are working assumptions based on general knowledge of radiology workflows. They should be confirmed with hospital staff before anyone relies on them.

Workflow and data flow



swimlane · 7 steps across EMR/HIS, RIS, modality and PACS

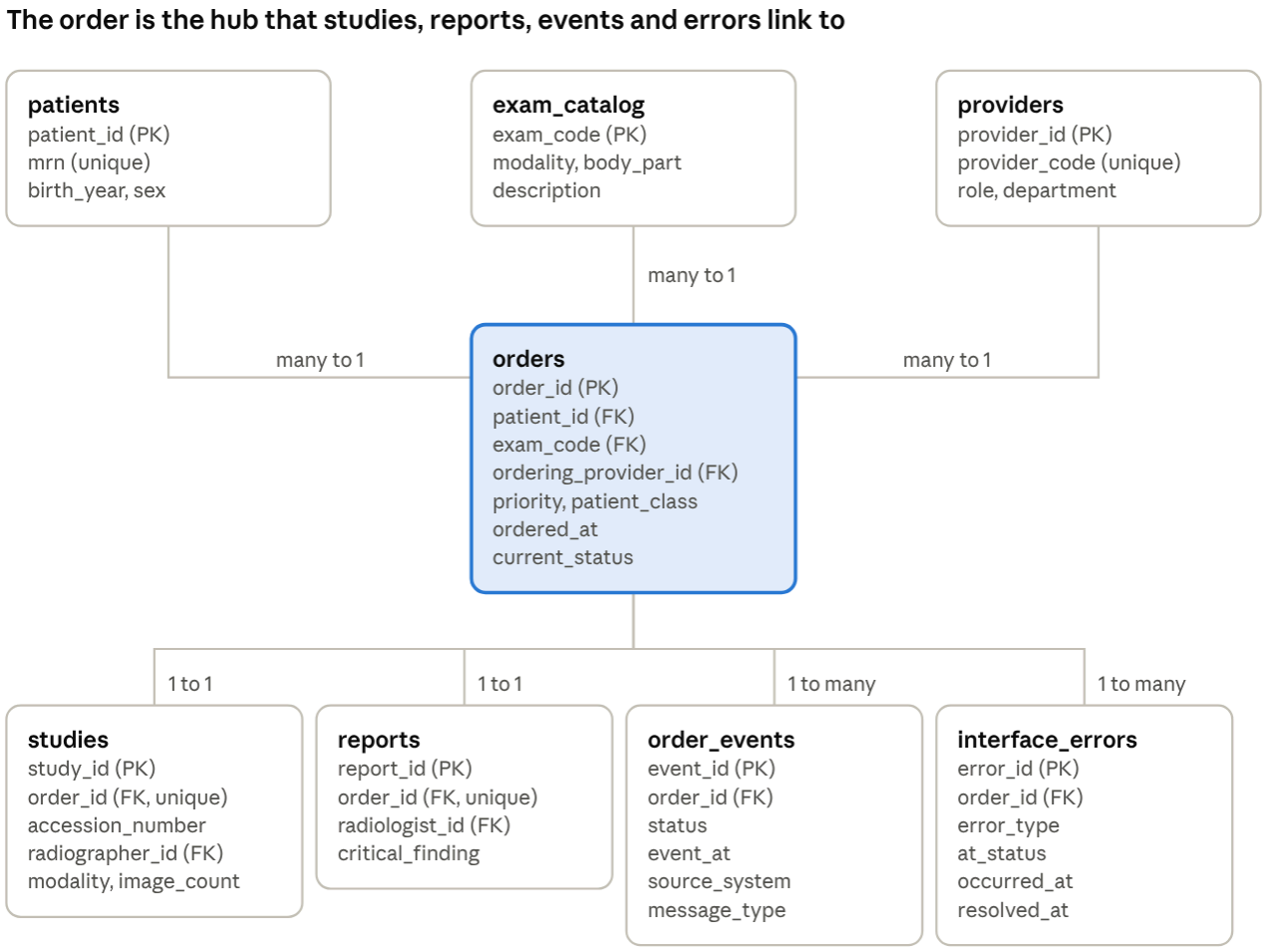
Each arrow is a message between systems (HL7 or DICOM). A, B and C mark the handoffs where the database logs interface errors.

Requirements

Functional requirements (FR) say what the systems must do at each handoff. Data requirements (DR) say what the database must guarantee so the path can be measured.

ID	Requirement	Why it matters
FR-1	The EMR sends each new imaging order to the RIS with patient identity, exam code, priority and clinical indication	A missing field forces a phone call and delays scheduling
FR-2	The RIS acknowledges every order, and the EMR shows the order as received	The doctor knows the order did not get lost
FR-3	The RIS publishes a modality worklist entry with correct patient data and an accession number	The radiographer scans the right patient, and the accession number later links images to the order
FR-4	The modality reports exam started and completed, then sends images to PACS	Start and end times can be measured, and images are not left on the device
FR-5	PACS links each study to its order by accession number and flags any mismatch	An unlinked study is invisible to the radiologist's queue
FR-6	The radiologist's worklist is sorted by priority and shows waiting time	STAT studies are read first, and long waits are visible
FR-7	A signed report goes back to the EMR as a result message, and the doctor is notified	The result reaches the person who acts on it
FR-8	Every interface failure is logged with the handoff, the time and the time it was fixed	IT can trace and fix problems, and the manager can see how often they happen
DR-1	Every status change is stored once per order with its time, source system and message type	One source of truth for all timing
DR-2	Timestamps in an order must be in the correct order, and no status may repeat	Bad data is caught by checks instead of reaching the dashboard
DR-3	A study has exactly one accession number, and a completed order has one study and one report	Prevents images or reports attached to the wrong order
DR-4	Service-level targets are stored by priority and kept separate from the data	Targets can change without rewriting history

Data model



entity relationship diagram · 8 tables

The `order_events` table is the single source of truth for time: one row per status change, with a unique key on order and status so no status repeats. Providers are also linked from studies (radiographer) and reports (radiologist). Check constraints limit statuses, priorities and error types, and a view turns the events into one row per order with the minutes spent in each stage. Illustrative service targets sit in a separate `kpi_targets` table.

Turnaround KPIs

On the synthetic data, the longest stage is the wait between images reaching PACS and the signed report: a median of 172 minutes and a 90th percentile of 560. Every other stage has a median under 30 minutes.

Priority	Orders	Median (min)	90th percentile (min)	Example target (min)	Within target
STAT	556	23	47	60	95.7%

Priority	Orders	Median (min)	90th percentile (min)	Example target (min)	Within target
URGENT	1,401	85	213	240	92.4%
ROUTINE	2,946	273	729	1,440	99.6%

Routine studies whose images arrive between 19:00 and 07:00 have a median report time of 762 minutes, against 255 by day. Interface errors touch 387 of 5,000 orders (7.7%): an order that did not reach the RIS (148 cases), a result not delivered to the EMR (144) and a patient ID mismatch in PACS (103).

These patterns come from assumptions written into the data generator, such as routine night studies waiting for the morning and 2 to 3 percent message failure at each handoff. They show that the queries find delays, not how any real hospital performs. The interactive dashboard shows the full breakdown by stage, shift, modality and error type. The code is on GitHub.

Limits and next steps

Limits of this version:

- All data is synthetic. Real hospital logs would need approval, de-identification and a data-sharing agreement.
- The stakeholder needs, targets and failure rates are assumptions to confirm with radiology staff.
- The model follows one clean path per order. It has no repeat exams, report addenda, system downtime or orders with several exams.

Next steps:

1. Walk this workflow through with a radiographer, a radiologist and a hospital IT analyst, and correct the steps and handoffs.
2. Replace the generator with a de-identified feed of real HL7 and DICOM status messages, keeping the same tables and queries.
3. Add an alert for orders that sit in one stage longer than its target, so delays are seen while they can still be fixed.